



KAHUNA

2023

Kahuna Project Report

A description of the Kahuna weather app

PRESENTED BY

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[GIT LINK](#)

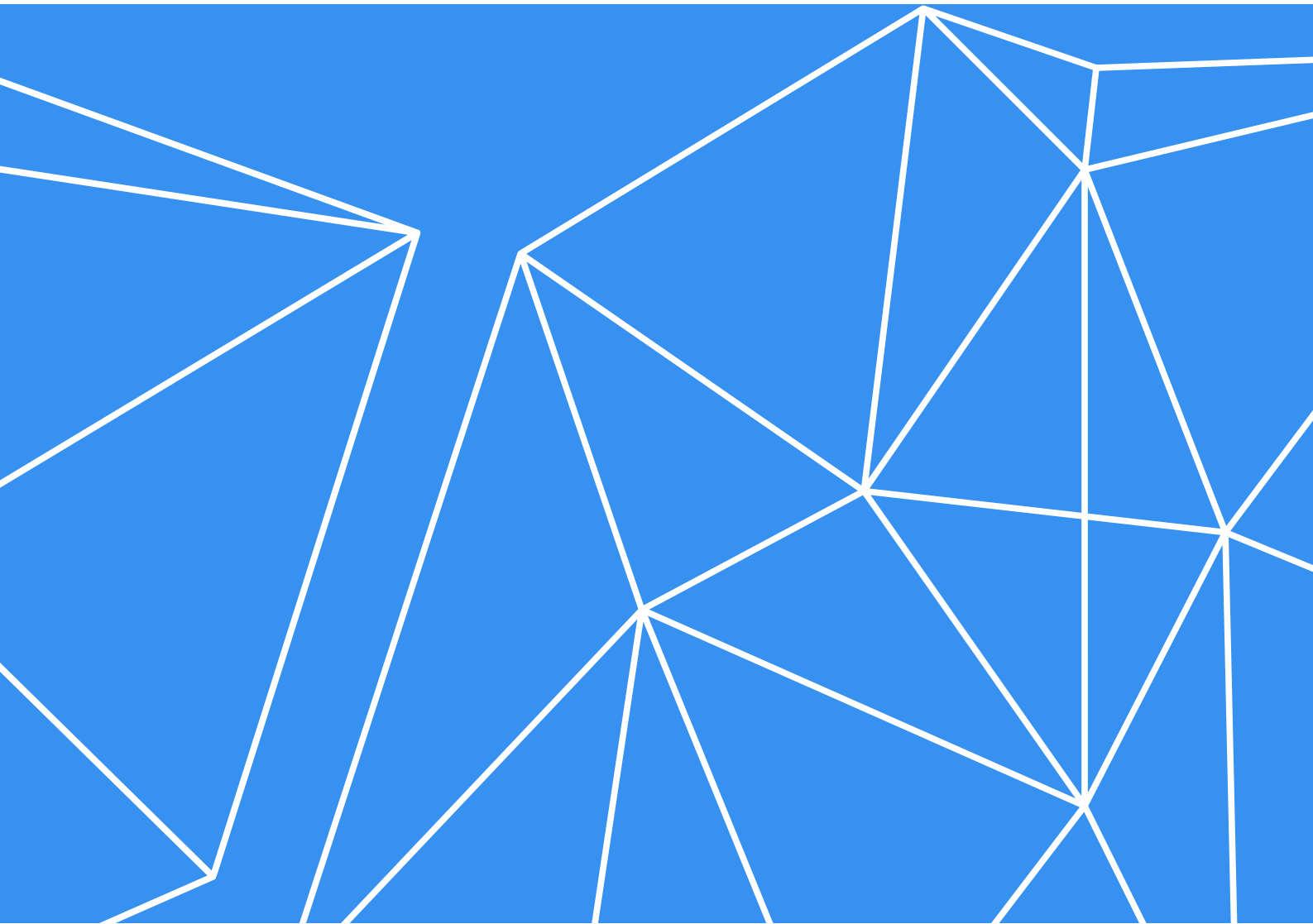


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Introduction

Kahuna

Kahuna is an innovative rain radar app designed specifically for surfers. With this app, surfers can now easily track weather patterns in real time and plan their surfing sessions accordingly. The app provides detailed information on rainfall intensity, wind direction, and wave height to ensure surfers are always prepared for the conditions.

Kahuna is the perfect app for those who want to stay ahead of the weather and maximize their time on the waves. It is user-friendly and intuitive, with a simple and easy-to-use interface that provides all the necessary information at a glance.

Kahuna is the perfect companion for your surfing adventures. With its accurate and up-to-date information, you can be confident in your ability to catch the best waves and enjoy the ultimate surfing experience.

Question

Weather App

We were assigned to create a weather app that took advantage of a weather API.

Criteria

A rain radar shows the amount of rain in mm/hour. But how much is 1mm of rain?

Do I need a towel to dry my hair when I'm inside? And does my sister need a towel?

How do I experience different amounts of rain?

I want an app that tells me how much it will rain in the next couple of hours.

How can an app tell me what it feels like to go outside in the rain?

How can we make a beginner-friendly surfing app?

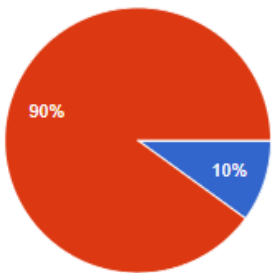
Methods

Survey

How old are you?

10 responses

 Copy



- <18
- 18-24
- 25-30
- 31+

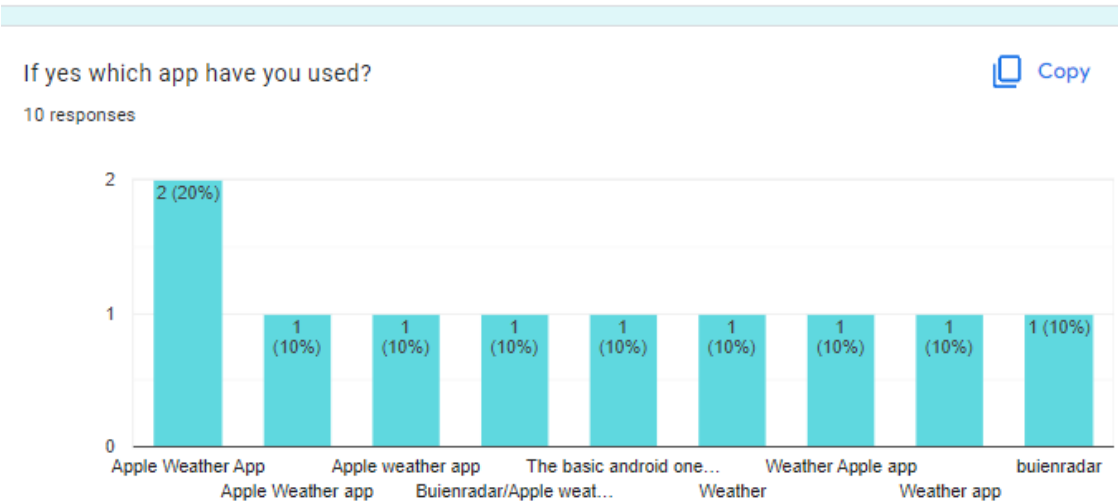
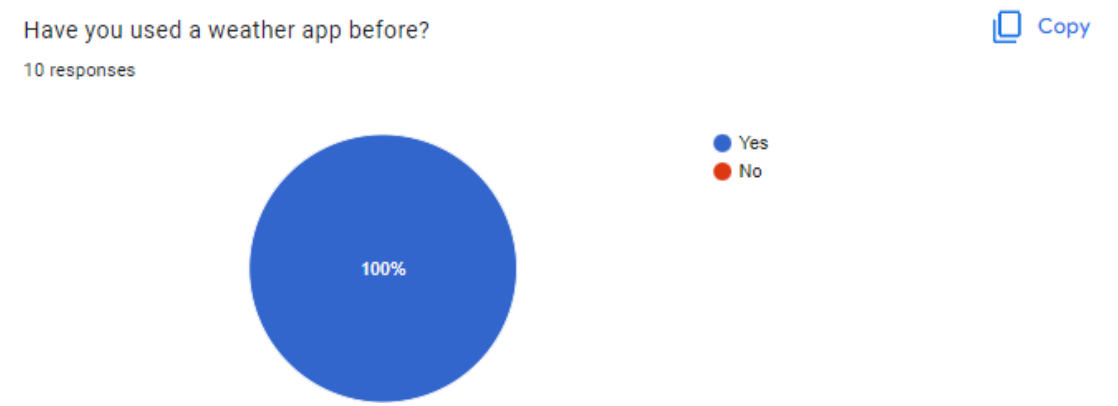
What is your work/study field?

10 responses

ICT
Catering
Smart Mobile
IT
High School
Community College Sierra
Sierra College
I am a student
I am working

Methods

Survey



Methods

Survey

What do you most look at on the app?

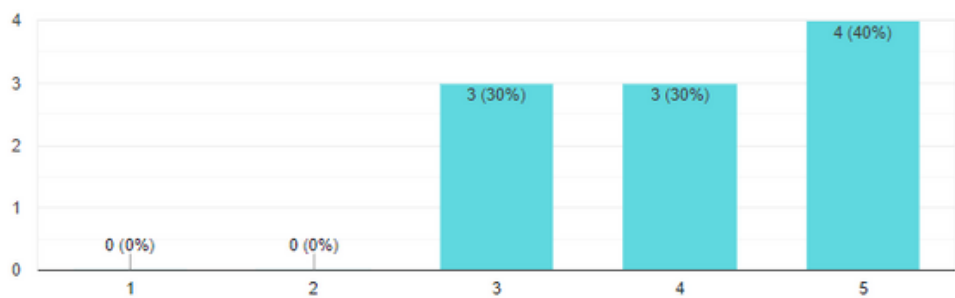
10 responses

Temp
temperature and if its raining or not
the weather its going to be throughout the day
the temperature
Temperature
Forcast
Weather/Rain
If it is going to rain
if it is going to be rainy today

How would you rate the app you use?

 Copy

10 responses



Methods

Survey

If possible what changes would you make to the app?

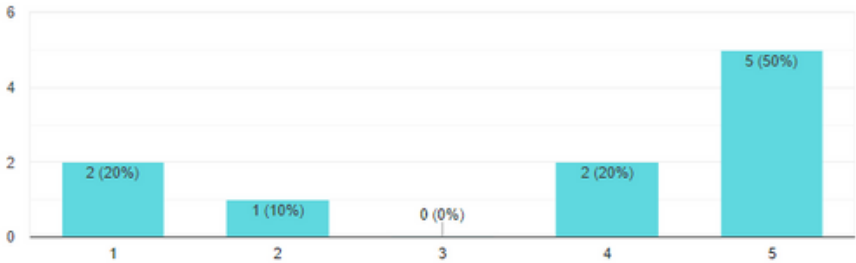
7 responses

easier to see other places temperature and wind speeds
Nothing its good and simple
Better design
add more options to it like: feels like temperature, and if it is gonna rain how much mm per square meters it will
The weather app on apple is not the most accurate, sometimes it rains even though it says it should not be raining
Better design on the buienradar
Better forecast

What is more important to you?

 Copy

10 responses

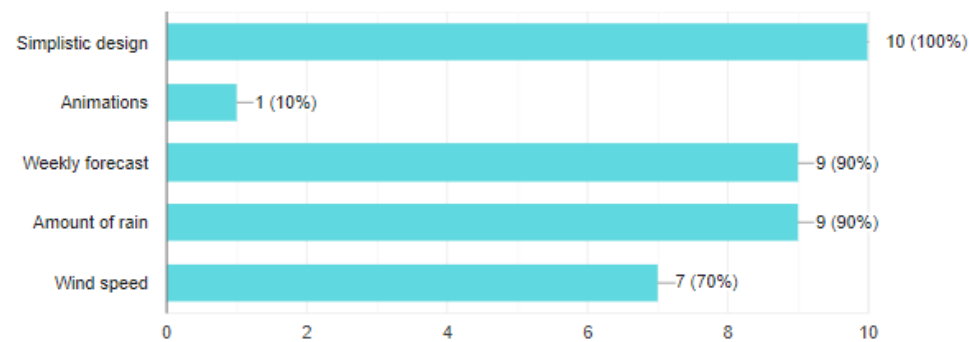


Methods

Survey

Which of these is essential for a good weather website in your opinion? [Copy](#)

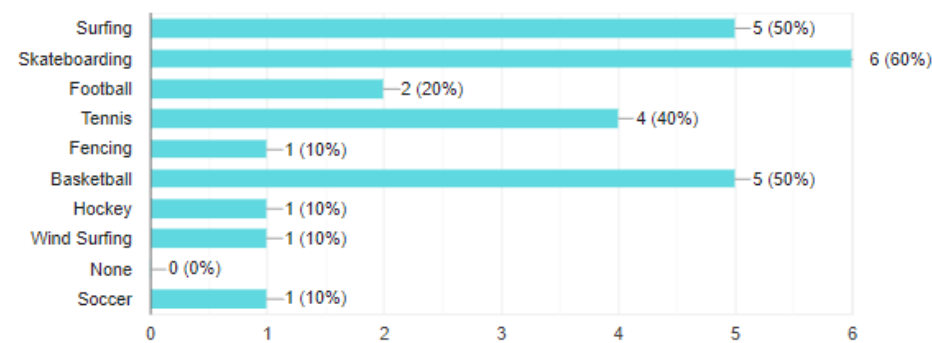
10 responses



Sports

What sports do you do? [Copy](#)

10 responses



Methods

Survey

If you surf or do any other outdoor activities what information would you need?

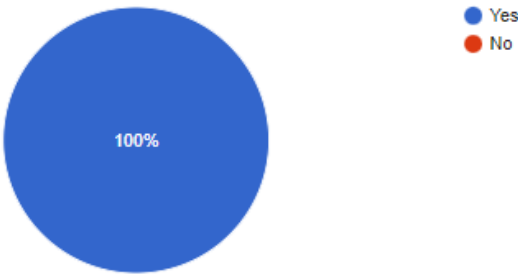
9 responses

Raining or not and how the waves are if surfing
temp and amount of rain
Rain or wind speed
temperature, wind information and rain information
Temp
Temp wind and rain
Rain/ Wind speed mostly
I think mostly the general weather information and wind speed
Displaying accurately if it is raining today and when

Would you use an app that shows the weather necessary to do sports outside?

 Copy

10 responses



Methods

Survey Results

From the survey we made, we can conclude that our target audience is 90% between the ages of 18-22 and 10% less than 18. Most of the users are currently students in university, and college and some of them are currently working. Out of all the answers we observed that all of them have used a Weather app or Rain radar app before. The apps they are currently using are mostly the built-in apps for Apple and Samsung and the minority is also using a rain radar app such as Buienradar. When we asked about what they mostly looked at in the app 50% of the answers were Temperature and the others were varying between the rain and the forecast for the week. The next question was about if the users are satisfied with the app they are using and the answers vary between 3 to 5 on a scale from 1 to 5. To conclude the users are currently satisfied with what they use but there could be improvements made. So that brings us to our next question which was what changes would you make to the app?

- Mostly we got responses about how the weather would feel even though you already have the temperature showing.
- Then we got responses about design as some people are not satisfied with the design of the app they are using and the Buienradar as well.
- And the minority of the left answers were about better forecasted, wind speed in other places and that they would not change anything.

Methods

Survey Results

Our next question was about information and readability and users showed that they would like to have better readability on the app as 50% of them answered that. Following the next question was about the essential parts of a good weather app/rain radar. The answers were pretty straightforward and we can observe that 100% of people want a simplistic design, 90% Weekly forecast, amount of rain, 70% wind speed, and 10% want animations. So in conclusion we should focus more on the simplistic design, forecast, and the amount of rain.

Here comes our next section of the survey which is sports. As we got information from the online research we decided that our app will be mainly focused on people who Surf, but as surfing is one of many sports we wanted to get information about what people in general who play sports want in an app. We started by asking what sports they play. And the answers were 60% skateboarding, 50% surfing and basketball, 40% tennis, 20% football and the rest were 10% with fencing, hockey, windsurfing, and soccer. Our next question was about the information they look at before they go out and do their outdoor activities 50% of the answers were about temperature and the other 50% were about rain and wind speed.

To conclude what we should show mostly in our app is the temperature, rain, and wind speed as our main points.

Methods

Competitor Analysis

Competitor Analysis Surfing App

Competitors	Weaknesses	Strengths	Link	Important Findings
Surfline	• Paid Cameras • No recommendations • Over	• Large database • Cameras for waves • Website & App • Good stats for surfing	https://www.surfline.com/	They have a paid model for the cameras on the beach.
Magicseaweed	• Super similar to Surfline • Paid Webcams	• Large Database • Webcams • App and website	https://magicseaweed.com/	Webcams and they sell clothing but it isn't targeted to help beginners.
Wisuki	• Dosent help surfers specifically • Targeted to any sport that needs wind. • Easy-to-use app and website • No Webcams	• Great stats about wind • App and website • Great for just checking the wind	https://nl.wisuki.com/	Too broad not much information about specific surfing. Great stats about the wind all over the world.
WSL(World Surfing Leauge)	• Hard to find what you are looking for • App is different than the site	• Targeted for more competition surfers • Good reputable stats • Good additional surfing news and videos	https://play.google.com/store/apps/details?id=com.wsl.android&hl=nl&gl=US	Great if you are a competition surfer or more interested in the surfing sport but definitely not beginner friendly.

Methods

Interview Questions

Main questions

- Age, Student/Job?
- Do you find the weather in the Netherlands frustrating?
- If you needed to know anything about the weather what would you need to know?
- Do you check the weather on any app or website before you go out?
- If yes what app or website do you use?
- If not then how do you check the weather? what do you look for?
- What do you like to look at before you leave?
- If you could change anything about your "weather app" what would you change?
- Do you use a weather radar?
- If yes why do you use it?
- If not why don't you use it?

Sports / Surfing

- Do you play any sports?
- Have you surfed before?
- Do you use anything to check the weather before you go out?
- If yes what do you look at and for before you go?
- Would you use a weather app for surfing?

Methods

Expert Interview

[INTERVIEW LINK](#)

[TRANSCRIPT LINK](#)

Interview Results

Expert Interview

- There is a market for a surfing weather app that targets beginners
- The main weather points we should look out for are. Wind Speed, Wave Height, Tide, and Swell
- Pieter and his friends use a variety of tools to predict the waves' conditions before going surfing, including weather reports, cameras, and websites like Surf Land and Magic Seaweed.

Methods

Interview

[INTERVIEW LINK](#)

[TRANSCRIPT LINK](#)

Interview Results

Gavin Interview

- Mentions that he would check the wind and tide levels before engaging in outdoor activities like surfing or playing golf, as they can have a significant impact on his performance.
- Would be interested in using a weather app that provides more in-depth information, such as wind speed and tide levels, if it were available
- Uses a weather widget on his phone's home screen to check the weather

Methods

Interview

[DIMITAR INTERVIEW LINK](#) [INTERVIEW SCRIPT](#)

Interview Results

Dimitar Interview

- Mentions that he finds the weather in the Netherlands frustrating, mainly looks at temperature of the day.
- Uses mostly the weather app that comes up with his android phone so he does not really use a rain radar app.
- Would use a rain radar app and thinks the overall idea is really useful.

Methods

POV & HMW

HMW

How might we create a good representation of the weather?

How might we make a more usable app for surfers of all skills?

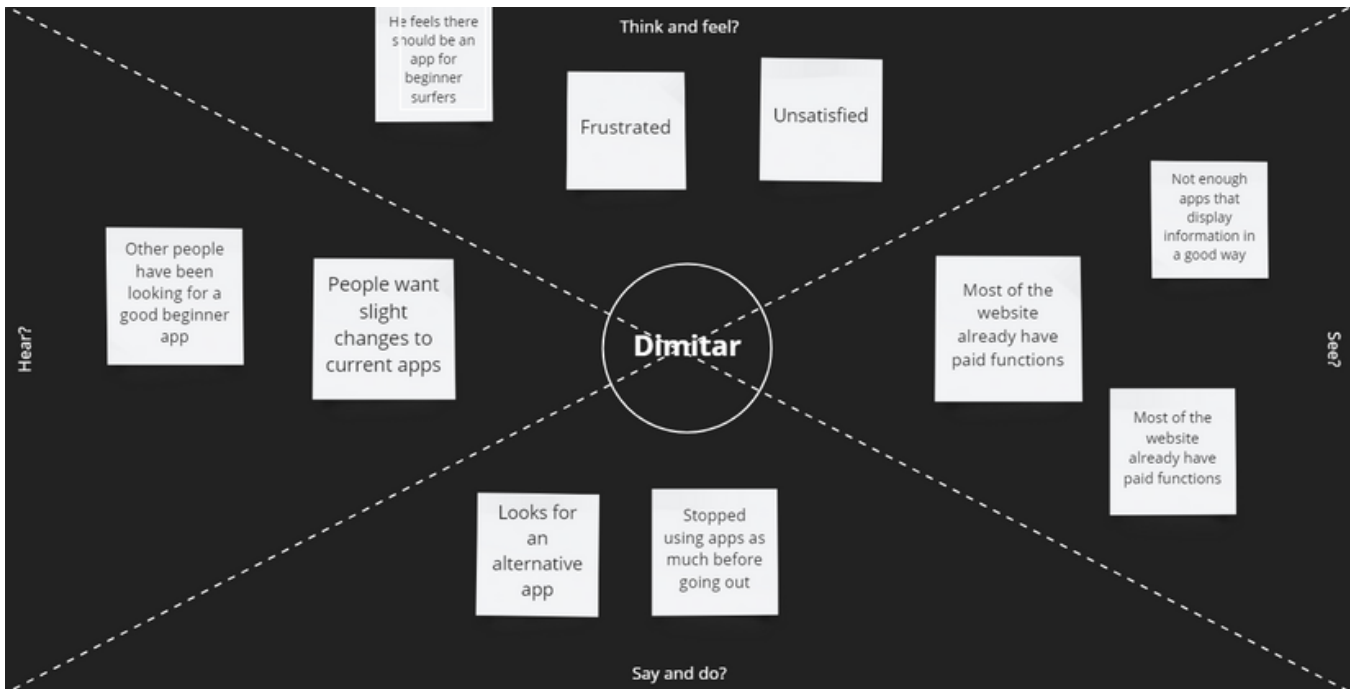
How might we improve upon existing weather apps and radars to make a unique and usable app?

POV

Users	Need	Insight
A surfer who needs weather insight before heading out.	To show accurate weather information for surfers.	Users need a precise and easy-to-use app that can allow surfers to see beaches, weather, and waves around the world quickly and for free.

Methods

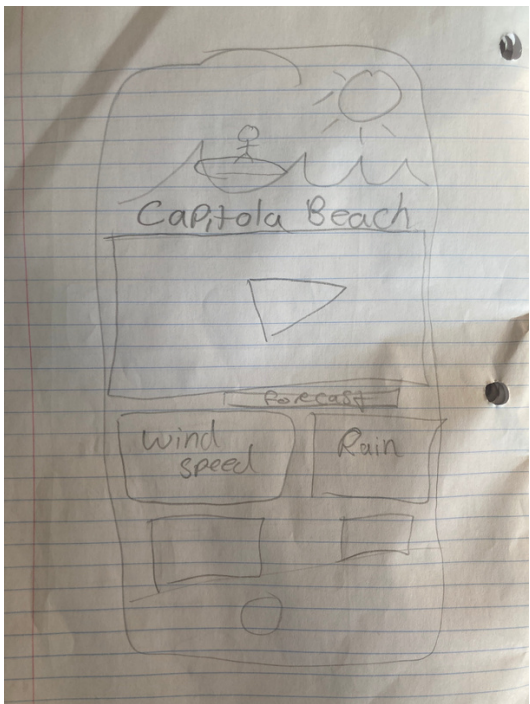
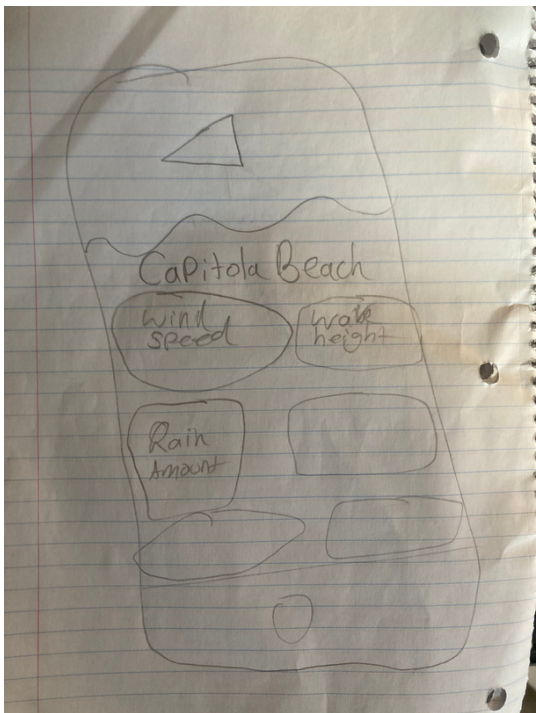
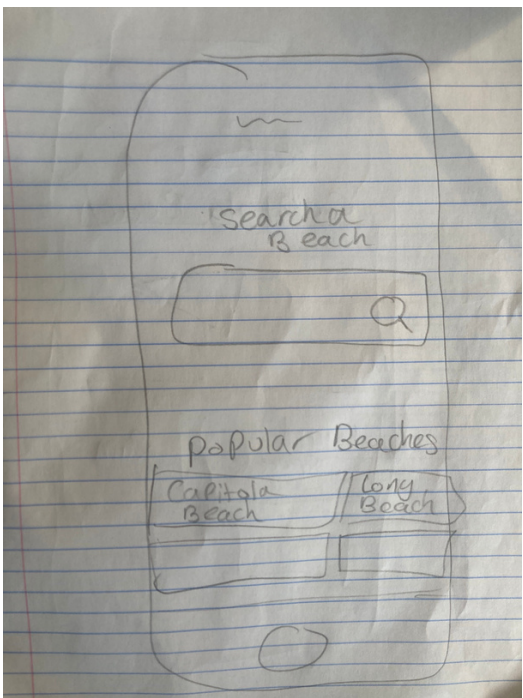
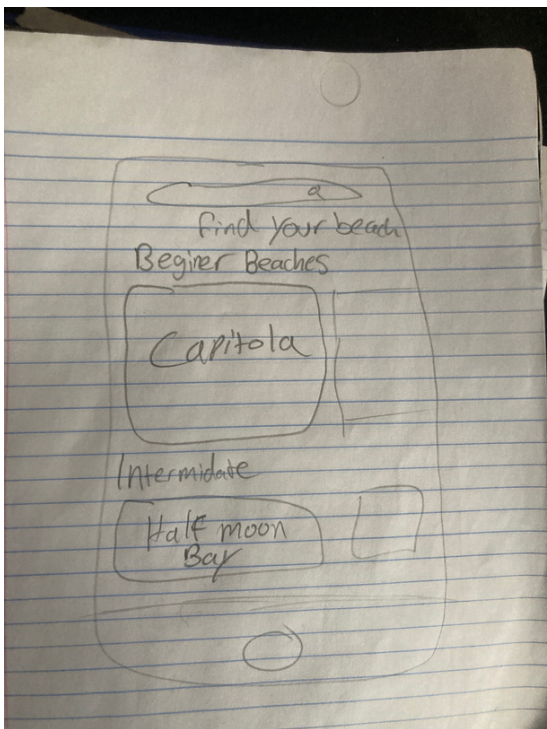
Empathy Map



The empathy map was done to capture the user's feelings and what he thinks about current apps. We used the information we got from the interviews and put it into an empathy map to conclude what changes need to be made in our app to satisfy the users more.

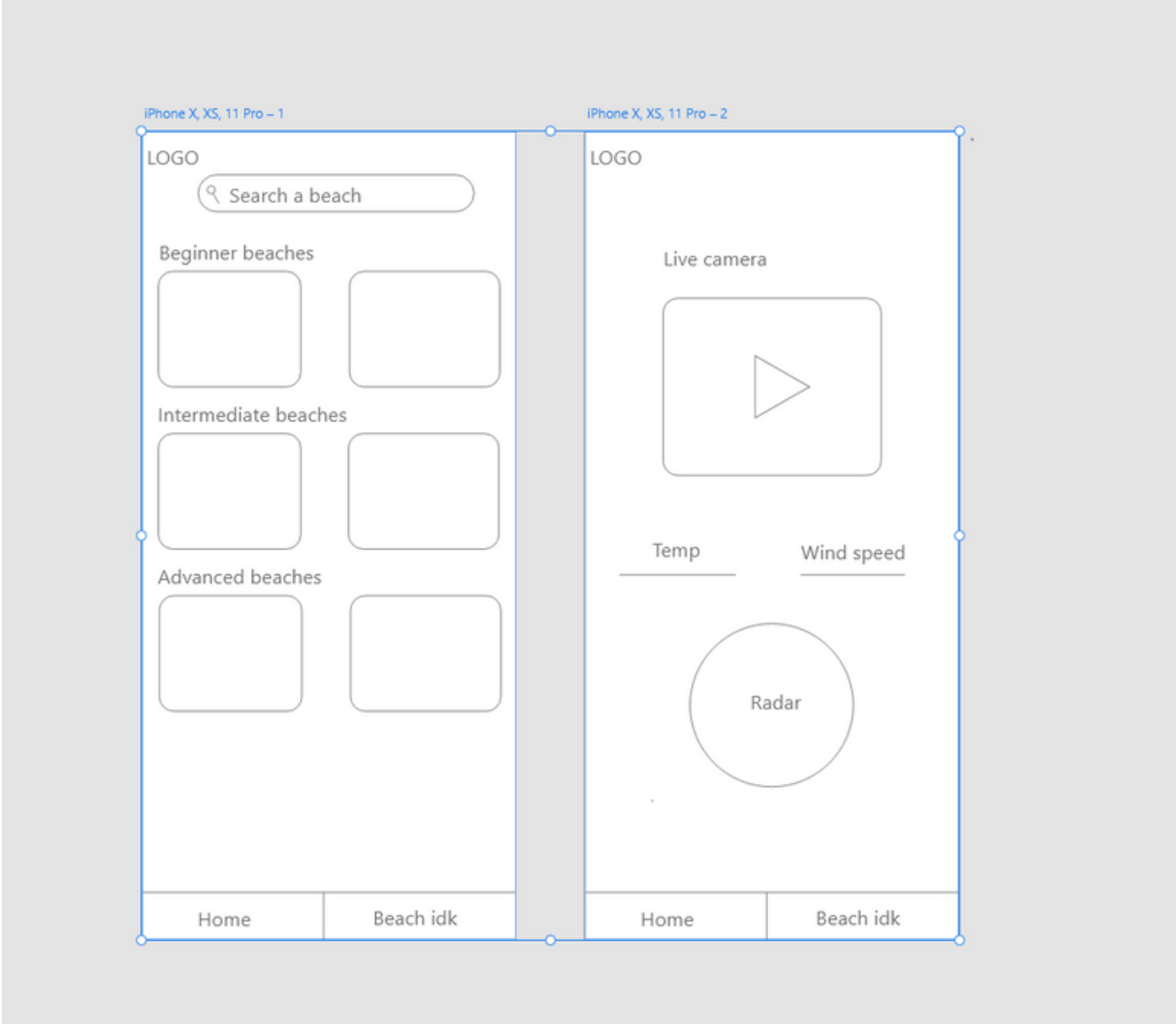
Methods

Paper Prototype



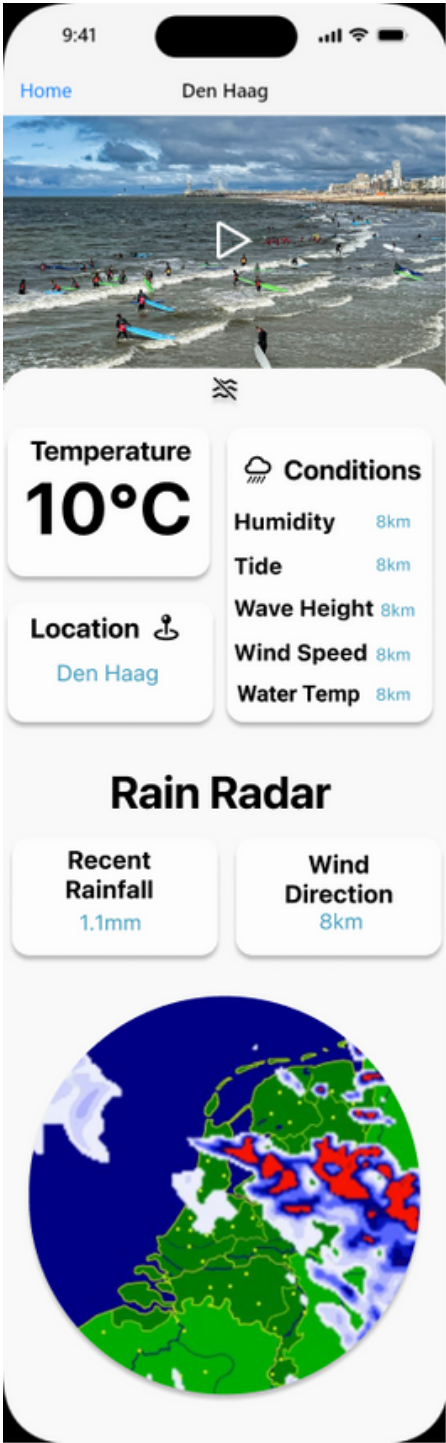
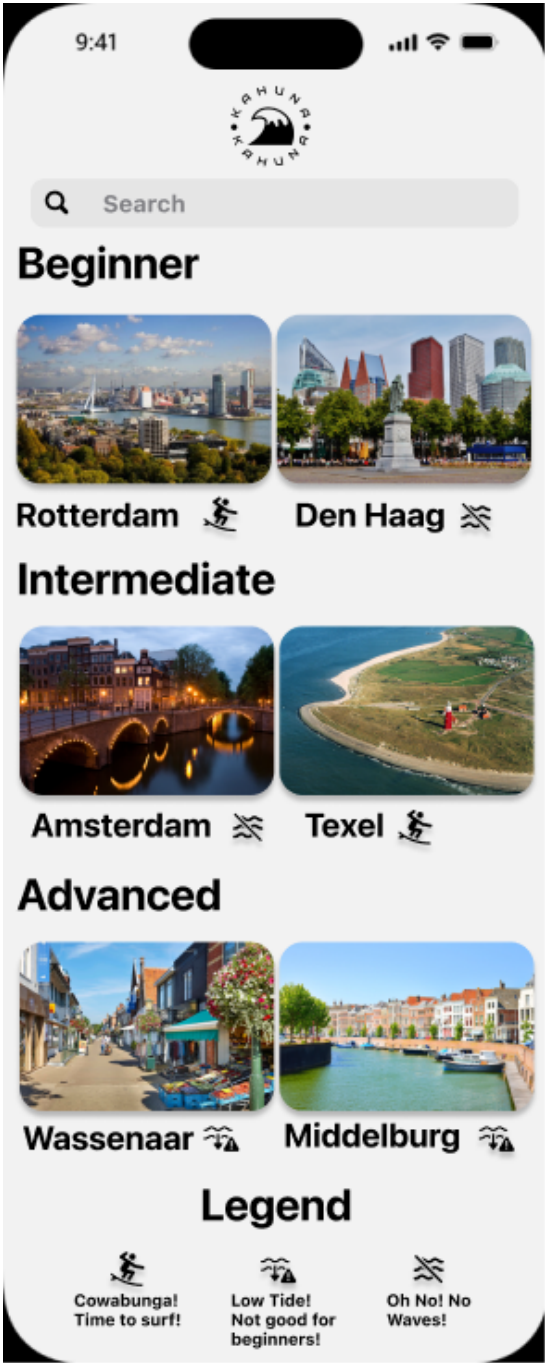
Methods

Low Fidelity Prototype



Methods

High Fidelity Prototype



Methods

Low-Fidelity User Testing

LINK TO USER TESTING

Conclusion of the first user testing.

The tester said that the app overall looked nice, as he was given the certain tasks he managed to complete which indicates that it is not hard to find information and the layout is good.

The feedback was mainly about how the face icons could be something more surfing and wave related but overall we had a good start and we could defiantly expand on our ideas.

Methods

High-Fidelity User Testing

LINK TO USER TESTING

Conclusion of the first high-fidelity user testing.

The tester said that the app overall looked nice, as he was given the certain tasks he managed to complete which indicates that it is not hard to find information and the layout is good.

The feedback we received about changing things was the blue boxes that had text on them and after reading apple guidelines that is true that they indicate a button. He also gave feedback about adding more icons on the details page, which we did.

LINK TO USER TESTING

Conclusion of the final user testing.

The tester was able to navigate the site and understood the layout well, the only things that would need improving are adding more color to the icons to indicate if its good to surf, and we need the legend in a more visible area since it is at the bottom of the screen is hard to see at times.

Results

[GIT LINK](#)

```
func fetchTemperature() {
    guard let city = selectedCity else {
        print("No city selected")
        return
    }

    guard let url = URL(string:
        "https://api.openweathermap.org/data/2.5/weather?lat=\(city.latitude)&lon=\(city
        .longitude)&units=metric&appid=8c714448edc2ce49f32ce5608cebf590") else {
        print("Invalid URL")
        return
    }

    let task = URLSession.shared.dataTask(with: url) { data, response, error in
        if let error = error {
            print("Error: \(error.localizedDescription)")
            return
        }
        guard let data = data else {
            print("Error: Data is nil")
            return
        }

        print(String(data: data, encoding: .utf8) ?? "")

        do {
            let decoder = JSONDecoder()
            let weatherData = try decoder.decode(WeatherData.self, from: data)
            DispatchQueue.main.async {
                self.temperature = "\(weatherData.main.temp)"
                self.windSpeed = "\(weatherData.wind.speed)"
                self.humidity = "\(weatherData.main.humidity)"
                self.windGust = "\(weatherData.wind.gust ?? 0.0)"
            }
        } catch {
            print("Error: \(error.localizedDescription)")
        }
    }

    task.resume()
}
```

The code uses the OpenWeather API to fetch information it. We are using Current Weather Data Collection and we have created structs accordingly to the api in which we take the information then save them in strings which we use in the app.

```
Text("Direction")
    .font(.title3)
    .fontWeight(.medium)
Image(systemName: "location.north.line")
    .foregroundColor(.blue)
    .rotationEffect(.degrees(Double(windDegree) ?? 0))
```

The code changes the direction of an arrow which shows the direction of the clouds.

Results

[GIT LINK](#)

```
var body: some View {
    VStack {
        if let image = radarImage {
            Image(uiImage: image)
                .resizable()
                .clipShape(Circle())
                .frame(maxWidth: 300, maxHeight: 340, alignment: .center)
        } else {
            Text("Loading...")
        }
    }
    .onAppear {
        fetchRadarImage()
    }
    .onReceive(timer) { _ in
        fetchRadarImage()
    }
}

private func fetchRadarImage() {
    guard let url = URL(string:
        "https://api.buienradar.nl/image/1.0/RadarMapNL?w=500&h=512&nocache=\(UUID()
        .uuidString)") else { return }
    URLSession.shared.dataTask(with: url) { data, response, error in
        guard let data = data, error == nil else {
            print("Error fetching radar image: \(error?.localizedDescription ??
                "Unknown error")")
            return
        }
        DispatchQueue.main.async {
            self.radarImage = UIImage(data: data)
        }
    }.resume()
}
```

This view represents another feature in the app which is the radar. We used Buienradar API for it. The radar updates every 5 minutes.

```
struct WebView: UIViewRepresentable {
    let url: URL

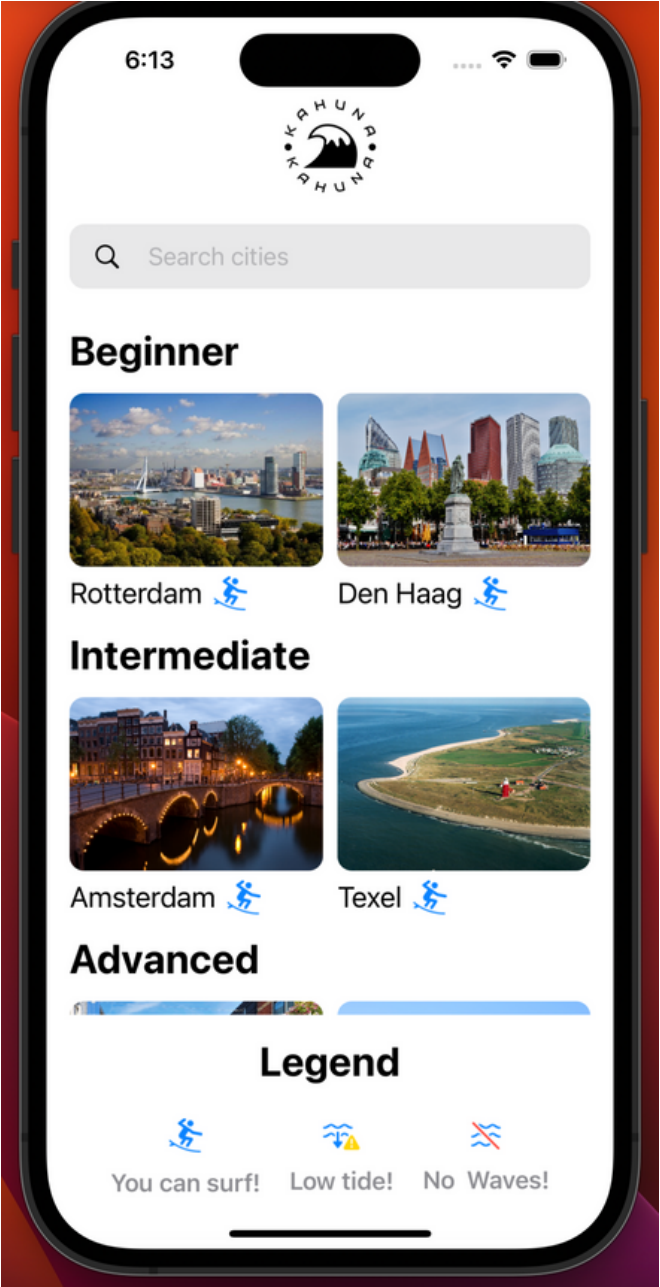
    func makeUIView(context: Context) -> WKWebView {
        let webView = WKWebView()
        webView.load(URLRequest(url: url))
        return webView
    }

    func updateUIView(_ webView: WKWebView, context: Context) {}
}

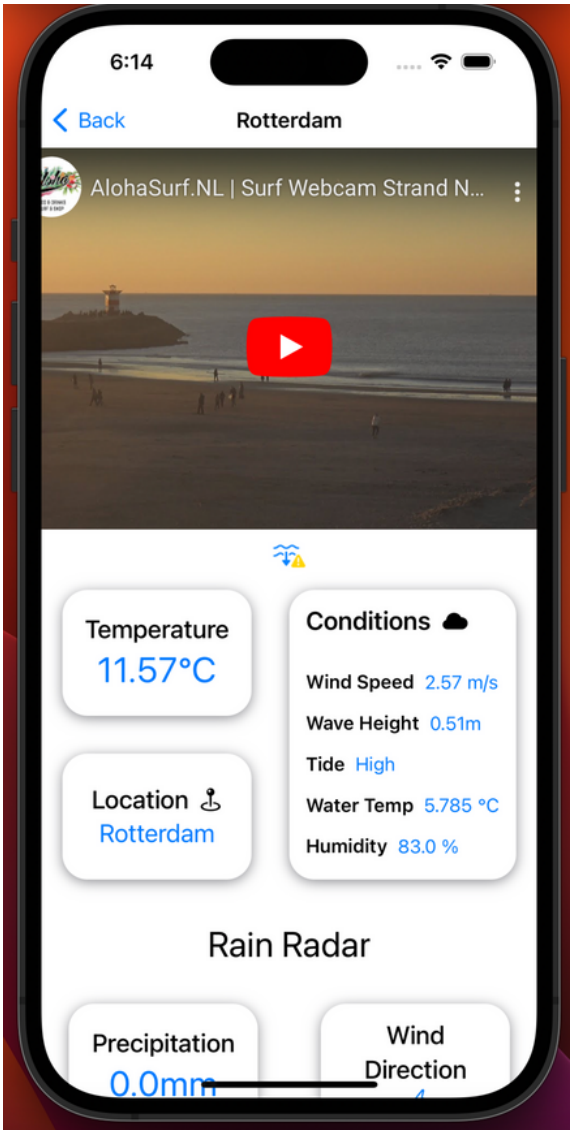
struct YoutubePlayerView: View {
    var body: some View {
        WebView(url: URL(string: "https://www.youtube.com/embed/EHK0x5QJEj4?controls=1&showinfo=0")!)
            .frame(width: 560, height: 315)
    }
}
```

This view shows the livestreaming feature we have it uses the webkit framework and takes a link.

Results



Results



Conculsion

In conclusion, we are overall satisfied with the work that we did. We did create a functioning weather app, there are only a few small little things that we would do differently. We successfully connected an API and a live stream to a Swift project and we found a good target to make our app unique. We can see that there is a good future for our product and a lot that we can expand and grow from.

There are a few small things we could do differently, we would have benefited from finding another API that has more wave and surfing targets so we could make all of our data reactive, if we had begun playing around with APIs more we might have been able to find one but overall it was successful. Additionally, during our showcase day, we got a lot of feedback about our Legend bar looking like a nav bar, this is a simple fix and something that would improve our usability.

Overall we challenged ourselves and were able to achieve what we set out to do, we learned a lot about how to program and design with Swift and mobile in mind.



Advice to Stakeholder

Overall maintaining our product is not difficult, we make use of OpenWeather's API. To expand and allow more widespread and more beach options we need more data, another API would be needed because OpenWeather does not have enough data about waves to have all the data. Most of the data in the CityView is accurate and reactive data but wave height and tide need a more focused API to be usable as a real product.

In the future, we would like to expand to other countries and beaches all over the world, once there is a more focused API we can use we only would need to have a connection with some of the restaurants and bars near the beach so we can get a live stream of the waves.

To monetize our product we are trying to make connections with existing surf schools and businesses near beaches around the world. We will sell the Livestream spot as advertising for local businesses along the beaches we are tracking.

